

Learning Outcomes and Assessment Map — RAG

Beyond the Demo

Learning Outcomes

By the end of this session, trainees will be able to:

1. **Explain** why hybrid retrieval (dense + keyword) combined via Reciprocal Rank Fusion is used instead of a single retrieval method or a weighted score average.
2. **Identify** the specific pipeline stage (ingestion, chunking, retrieval, or generation) responsible for a given incorrect or ungrounded answer.
3. **Analyze** a retrieval score distribution to determine whether a refusal threshold is correctly calibrated, using real evaluation data rather than assumption.
4. **Diagnose** whether a wrong-document retrieval is caused by ambiguous corpus content (e.g. near-identical boilerplate across documents) versus a genuine retrieval-ranking defect.
5. **Evaluate** whether a citation returned by the system actually supports the generated answer, distinguishing "a citation exists" from "the answer is grounded in it."
6. **Compare** direct and indirect prompt injection risks in a RAG system, and explain why retrieved content must be treated as data rather than instructions.

Assessment Map

Learning Outcome	Assessment Method	Evidence of Learning
LO1 — Explain RRF	Formative Q&A during lecture	Trainee correctly explains why RRF avoids the score-comparability problem
LO2 — Identify failure stage	Practical lab assessment	Correct diagnosis submitted in Lab Task 5's written reasoning
LO3 — Analyze threshold calibration	Stretch challenge assessment	Correct hand-computed max-score comparison across two rrf_k values
LO4 — Diagnose wrong-document retrieval	Practical lab assessment	Trainee correctly identifies whether observed retrieval

		reproduces the boilerplate-collision pattern
LO5 — Evaluate citation groundedness	Formative + practical	Trainee checks all 7 citation fields are populated and plausible, not just present
LO6 — Compare injection risks	Stretch challenge assessment	Trainee's injection test report correctly identifies whether the instruction was followed and why

Assessment Types Used

- ❑ **Formative assessment** — in-session Q&A during Slides checking understanding before the lab begins.
- ❑ **Practical lab assessment** — Tasks 1–5, each with an expected-output property to verify, graded on whether the trainee's actual observation matches the expected property (not an exact string).
- ❑ **Stretch challenge assessment** — the 3 stretch challenges, each requiring a short written analysis in addition to the executed change, since the objective is reasoning, not just running a command.
- ❑ **Optional final assessment** — a written short-answer question: "You observe a claim adjudication that states an incorrect deductible. Using this project's architecture, describe the exact sequence of checks you would perform to determine whether the defect is in ingestion, chunking, retrieval, or the deterministic calculate_payout tool." This tests LO2 and LO4 together without requiring live repository access.